POKHARA UNIVERSITY

Level: Bachelor | Semester: Spring | Year : 2018
Programme: BE | Full Marks: 100
Course: Computer Architecture | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) What are the characteristics of computer system? Draw and explain the basic computer organization in detail. 8
   b) What is RTL? Explain it when data transfer takes place between one and four bit register. 7
2. a) How a floating point number is represented in computer system? Verify the operation (9) x (-2) using signed magnitude data 8
   b) What is sequencer? Draw and explain the micro programmed control unit in detail. 7
3. a) What is instruction cycle? Write the necessary micro-operations for fetch, decode, indirect and interrupt cycles. 8
   b) What is mapping? What are the different cache memory mapping technique? 7
4. a) What is DMA? Explain why DMA is used instead of interrupt driven and programmed I/O? 7
   b) What is instruction pipeline? Explain 3-segment instruction pipelining used in RISC computer. 8
5. a) How a computer system can be classified according to flyn's? Design an arithmetic pipeline to speed up the computer to solve the expression $A_i * C_i + D_i$ for i=0, 1,2,3,4. 8
   b) What are the function of ISR? Explain how interrupt are processed in computer system. 7
6. a) How floating point number can be added or subtracted using pipelining techniques? 8
   b) What are different interconnection structure when there are multiple 7

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CPUs?

7. Write short notes on: (Any two)
   a) Alternative Chip Organization
   b) Types of registers
   c) Control memory organization and mapping logic

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